Matthew Diep

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EDUCATION

University of California, Berkeley **Exp. Dec 2026**

B.A. Data Science; B.A. Economics

- Relevant Coursework: Data Engineering, Data Analysis and Regression, Machine Learning, Data Structures and Algorithms, Computational Programming, Probability, Statistical Inference in R, Econometrics, Game Theory

SKILLS

Languages: Python, SQL, Java, R, JavaScript
Data & Analytics: Snowflake, dbt, Tableau, Kimball Dimensional Modeling, Pandas, NumPy, Scikit-learn, Matplotlib
Tools & Design: Git, GitHub, Jupyter, Microsoft Excel, Microsoft Power Automate, UI/UX Design

EXPERIENCE

Data Platform Engineering Intern, *Wind River Systems* **May 2026 – Present**

- Modeled an end-to-end dbt pipeline on Snowflake for contract renewal and opportunity data, using surrogate keys, joins, and aggregations, and defined dbt exposures linking 50+ production workbooks to upstream models for lineage visibility.
- Implementing an automated system test that mirrors the Tableau production environment and validates workbooks against test-stage data sources, with Power Automate Teams alerts and an HTML findings summary before each release.
- Developed a Python desktop app that automates Tableau asset promotion across three deployment stages with GitHub version control, cutting each migration from 2 hours to 10 minutes (92%) and eliminating manual deployment errors.
- Built standalone Python tools that sync dbt metadata with Tableau .tds files, with in-app description editing, and export field-level lineage, keeping definitions consistent and showing change impac**t** before production release.
- Designed and launched the website and registration for Tableau Day, a company-wide event for Wind River and Aptiv.

Venture Capital Analyst Intern, *In Collaboration with Ignite XL* **Aug 2024 – Sept 2024**

- Evaluated 100+ pre-seed and seed-stage startups across multiple industries and authored investment briefs and one-page memos that synthesized market and quantitative data to support shortlisting **decisions**, presenting one to Ignite XL.

Co-Founder, *Exceptional STEAM Nonprofit Organization* **Aug 2019 – Aug 2023**

- Founded a nonprofit to address limited STEAM access, delivering 9-week camps for 30+ Title I students in the U.S. and 12 in Taiwan and India, broadening access to STEAM education across local and international communities.
- Designed an interactive curriculum with weekly experiments in physics, chemistry, math, and engineering, and built virtual lab simulations on www.exceptionalsteam.org to expand international accessibility, boosting site usage by 60%.

PROJECTS

Feynman - AI Voice Learning Agent *(Claude, Next.js, Deepgram, Redis, Voyage AI, Vercel)* **June 2026**

- Engineered a full-stack, voice-first learning agent that builds a personalized knowledge graph from user notes and evaluates spoken explanations in real time using Claude, scoring concept mastery and surfacing knowledge gaps.
- Architected a multi-API pipeline integrating Deepgram STT/TTS for voice I/O, Voyage AI for concept embeddings, and Redis vector search (KNN) to retrieve semantically related nodes from the knowledge graph at inference time.
- Implemented persistent memory across sessions in Redis, storing mastery scores, misconceptions, and learning preferences to enable the agent to adapt its feedback to each user's evolving learning profile.

Waze Churn Prediction & User Behavior Analysis *(Python, Pandas, Seaborn, Scikit-learn, Jupyter)* **July 2025**

- Analyzed 20,000+ Waze user records, using two-sample t-tests to compare device types and surface churn indicators.
- Developed predictive models (Logistic Regression, Random Forest, XGBoost) on 2,800+ churn cases; the model
- achieved 81% accuracy with strong recall, supporting early intervention strategies for user retention.

Save Dinos - Gamified Visa Wallet *(JavaScript, Next.js, MongoDB, Tailwind CSS)* **October 2025**

- Built a Next.js and MongoDB student wallet that gamifies budgeting through evolving dinosaur goals, leveling rewards, social streaks, and badges, featuring a clean, responsive Tailwind UI designed to boost engagement.
- Integrated Visa Developer Platform with mutual TLS, MCC-based spend controls, and merchant offers; executed 100+ end-to-end Visa sandbox interactions while separating payment flows from app data for security and scale.